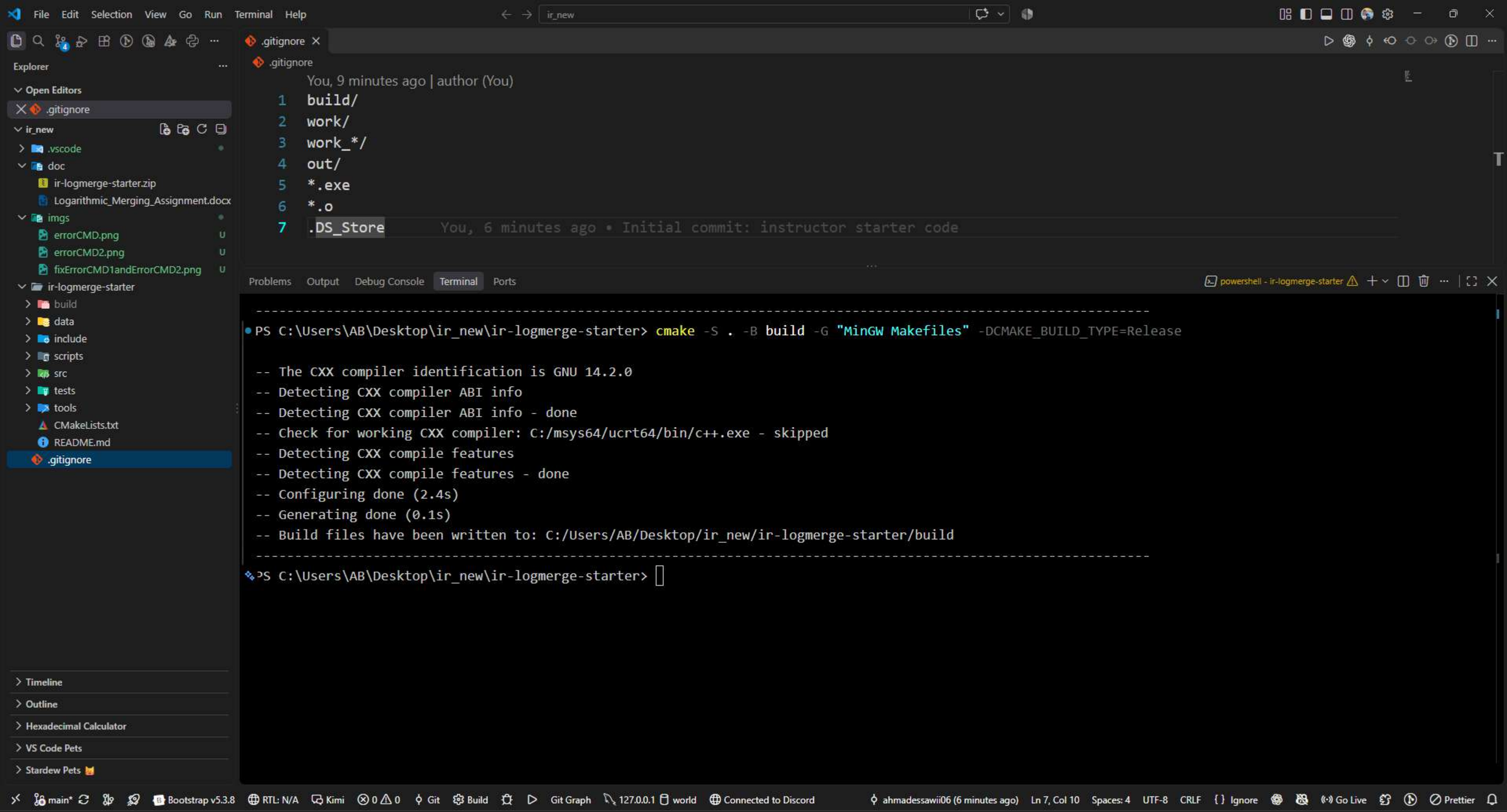

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Stardew Pets
.gitignore
You, 9 minutes ago | author (You)
1 build/
2 work/
3 work_*/
4 out/
5 *.exe
6 *.o
7 .DS_Store You, 6 minutes ago • Initial commit: instructor starter code
Problems Output Debug Console Terminal Ports
powershell - ir-logmerge-starter
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> cmake -S . -B build -G "MinGW Makefiles" -DCMAKE_BUILD_TYPE=Release
-- The CXX compiler identification is GNU 14.2.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/msys64/ucrt64/bin/c++.exe - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (2.4s)
-- Generating done (0.1s)
-- Build files have been written to: C:/Users/AB/Desktop/ir_new/ir-logmerge-starter/build
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter>
main* Bootstrap v5.3.8 RTL: N/A Kimi 0 0 Git Build Git Graph 127.0.0.1 world Connected to Discord ahmadessawii06 (6 minutes ago) Ln 7, Col 10 Spaces: 4 UTF-8 CRLF Ignore Go Live Prettier

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ir_new

.gitignore

.gitignore

You, 9 minutes ago | 1 author (You)

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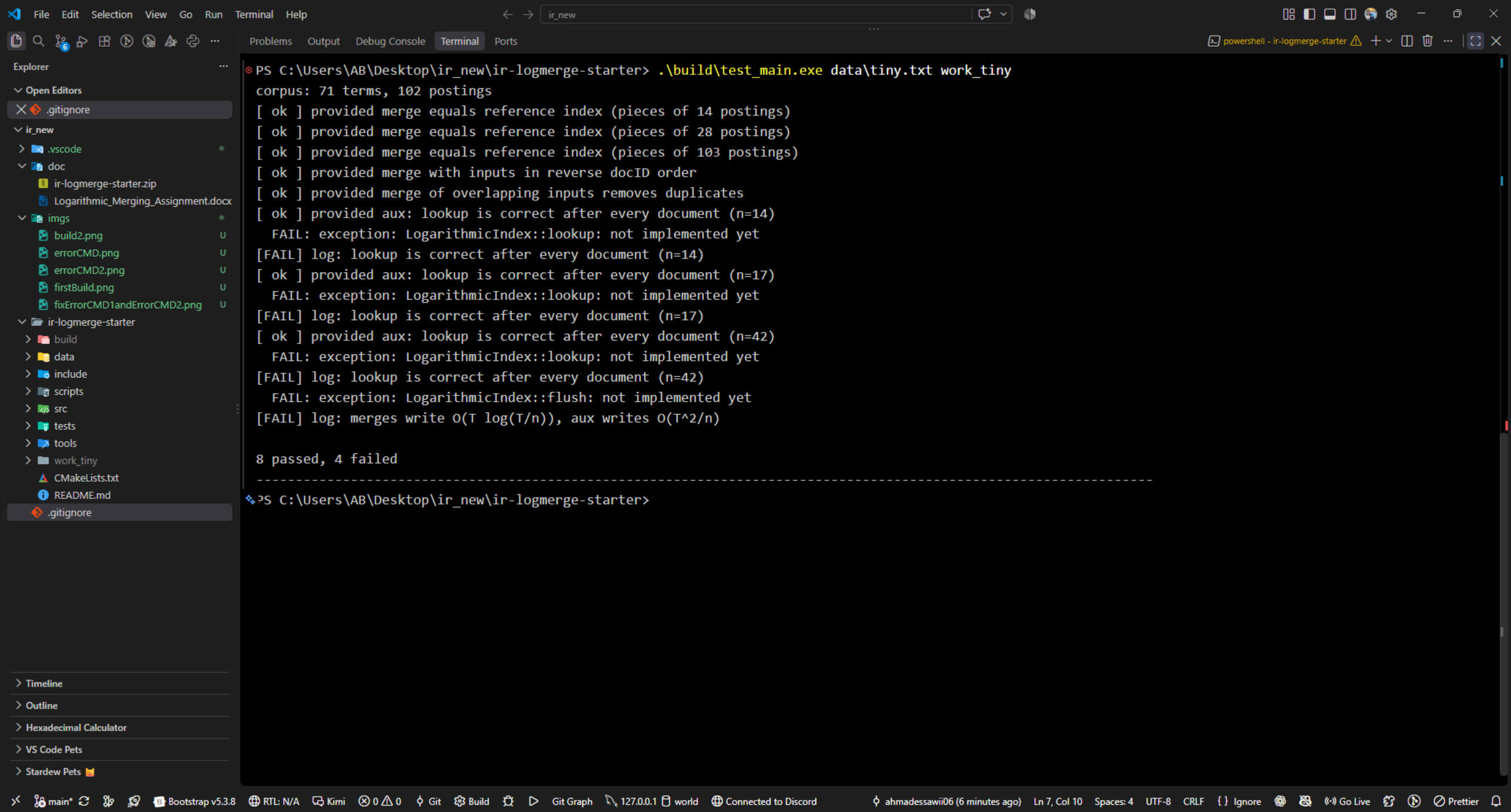
Problems Output Debug Console Terminal Ports

powershell - ir-logmerge-starter

```
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> cmake --build build -j
[  5%] Building CXX object CMakeFiles/irindex.dir/src/index_io.cpp.obj
[ 11%] Building CXX object CMakeFiles/irindex.dir/src/memory_index.cpp.obj
[ 17%] Building CXX object CMakeFiles/irindex.dir/src/dynamic_base.cpp.obj
[ 23%] Building CXX object CMakeFiles/irindex.dir/src/merge.cpp.obj
[ 29%] Building CXX object CMakeFiles/irindex.dir/src/aux_main.cpp.obj
[ 35%] Building CXX object CMakeFiles/irindex.dir/src/logarithmic.cpp.obj
[ 41%] Linking CXX static library libirindex.a
[ 41%] Built target irindex
[ 47%] Building CXX object CMakeFiles/build_dynamic.dir/tools/build_dynamic.cpp.obj
[ 52%] Building CXX object CMakeFiles/lookup.dir/tools/lookup.cpp.obj
[ 58%] Building CXX object CMakeFiles/verify.dir/tools/verify.cpp.obj
[ 64%] Building CXX object CMakeFiles/gen_corpus.dir/tools/gen_corpus.cpp.obj
[ 70%] Building CXX object CMakeFiles/test_main.dir/tests/test_main.cpp.obj
[ 76%] Linking CXX executable lookup.exe
[ 82%] Linking CXX executable verify.exe
[ 82%] Built target lookup
[ 88%] Linking CXX executable build_dynamic.exe
[ 88%] Built target verify
[ 94%] Linking CXX executable gen_corpus.exe
[ 94%] Built target build_dynamic
[ 94%] Built target gen_corpus
[100%] Linking CXX executable test_main.exe
[100%] Built target test_main
--------------------------------------------------------------------------------------------------------------------------------

PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter>
```

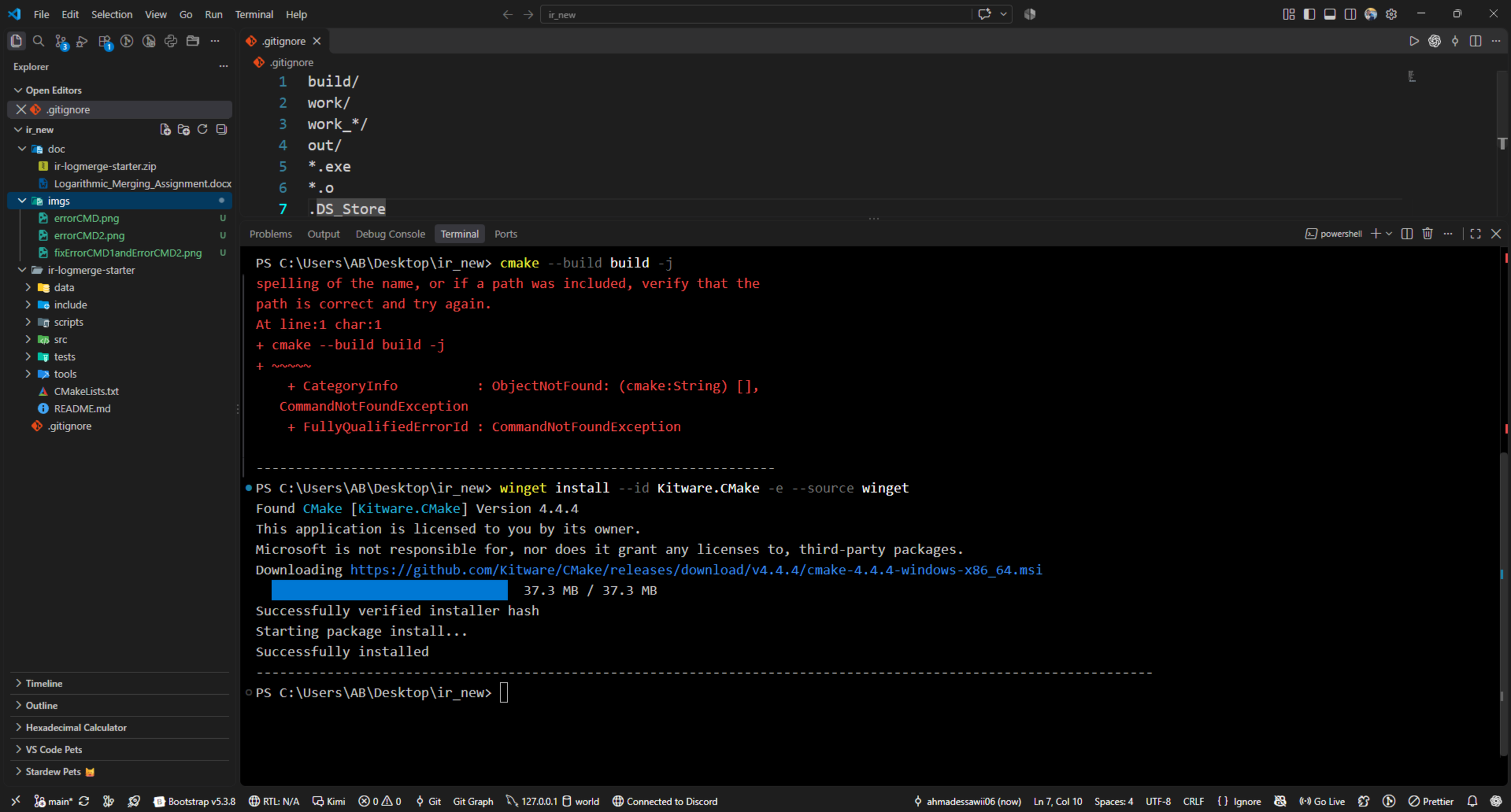
main* Bootstrap v5.3.8 RTL: N/A Kimi 0 0 Git Build Git Graph 127.0.0.1 world Connected to Discord ahmadessawii06 (6 minutes ago) Ln 7, Col 10 Spaces: 4 UTF-8 CRLF { } Ignore Go Live Prettier


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Hexadecimal Calculator
VS Code Pets
Stardew Pets
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> .\build\test_main.exe data\tiny.txt work_tiny
corpus: 71 terms, 102 postings
[ok] provided merge equals reference index (pieces of 14 postings)
[ok] provided merge equals reference index (pieces of 28 postings)
[ok] provided merge equals reference index (pieces of 103 postings)
[ok] provided merge with inputs in reverse docID order
[ok] provided merge of overlapping inputs removes duplicates
[ok] provided aux: lookup is correct after every document (n=14)
FAIL: exception: LogarithmicIndex::lookup: not implemented yet
[FAIL] log: lookup is correct after every document (n=14)
[ok] provided aux: lookup is correct after every document (n=17)
FAIL: exception: LogarithmicIndex::lookup: not implemented yet
[FAIL] log: lookup is correct after every document (n=17)
[ok] provided aux: lookup is correct after every document (n=42)
FAIL: exception: LogarithmicIndex::lookup: not implemented yet
[FAIL] log: lookup is correct after every document (n=42)
FAIL: exception: LogarithmicIndex::flush: not implemented yet
[FAIL] log: merges write O(T log(T/n)), aux writes O(T^2/n)
8 passed, 4 failed
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter>
main* Bootstrap v5.3.8 RTL: N/A Kimi 0 0 Git Build Git Graph 127.0.0.1 world Connected to Discord ahmadessawii06 (6 minutes ago) Ln 7, Col 10 Spaces: 4 UTF-8 CRLF { } Ignore Go Live Prettier

```
 ------------------------------------------------------------
PS C:\Users\AB\Desktop\ir_new> cmake -S . -B build -DCMAKE_BUILD_TYPE=Release

 cmake : The term 'cmake' is not recognized as the name of a
 cmdlet, function, script file, or operable program. Check the
 spelling of the name, or if a path was included, verify that the
 path is correct and try again.
 At line:1 char:1
 + cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
 + ~~~~~
     + CategoryInfo          : ObjectNotFound: (cmake:String) [],
    CommandNotFoundException
     + FullyQualifiedErrorId : CommandNotFoundException
```

```
PS C:\Users\AB\Desktop\ir_new> cmake --build build -j
cmake : The term 'cmake' is not recognized as the name of a
cmdlet, function, script file, or operable program. Check the
spelling of the name, or if a path was included, verify that the
path is correct and try again.
At line:1 char:1
+ cmake --build build -j
+ ~~~~~
    + CategoryInfo          : ObjectNotFound: (cmake:String) [],
   CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException
```


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.gitignore
1 build/
2 work/
3 work_*/
4 out/
5 *.exe
6 *.o
7 .DS_Store
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\AB\Desktop\ir_new> cmake --build build -j
spelling of the name, or if a path was included, verify that the
path is correct and try again.
At line:1 char:1
+ cmake --build build -j
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (cmake:String) [],
CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
PS C:\Users\AB\Desktop\ir_new> winget install --id Kitware.CMake -e --source winget
Found CMake [Kitware.CMake] Version 4.4.4
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/Kitware/CMake/releases/download/v4.4.4/cmake-4.4.4-windows-x86_64.msi
37.3 MB / 37.3 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\AB\Desktop\ir_new>
main* Bootstrap v5.3.8 RTL: N/A Kimi 0 0 Git Git Graph 127.0.0.1 world Connected to Discord
ahmadessawii06 (now) Ln 7, Col 10 Spaces: 4 UTF-8 CRLF { } Ignore Go Live Prettier

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ir_new
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run_experiments_local.ps1 U
run_experiments.sh
ir-logmerge-starter > scripts > run_experiments_local.ps1 > ...
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run_experiments.sh ir-logmerge-starter\sc...
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build1.png
build2.png
buildCMake_Before_Doning_Any_Thin...
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write_50k_doc.png
ir-logmerge-starter
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run_experiments_local.ps1 U
run_experiments.sh
src
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work_big
aux_141954
aux_141957
aux_425862
cost_aux
cost_log
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Stardew Pets
29 'mode,n,flushes,merges,max_disk_indexes,postings_written,postings_written_with_finalize,write_amplification,seconds' |
30 Set-Content -LiteralPath $summary -Encoding ASCII
31
5 references
32 function Get-Metric {
Problems Output Debug Console Terminal Ports
powershell - ir-logmerge-starter
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> powershell -NoProfile -ExecutionPolicy Bypass -File .\scripts\run_experiments_local.ps1
Running aux_400000
OK: 50000 terms match the reference index
Running log_50000
OK: 50000 terms match the reference index
Running log_100000
OK: 50000 terms match the reference index
Running log_200000
OK: 50000 terms match the reference index
Running log_400000
OK: 50000 terms match the reference index
Final indexes retained in: C:\Users\AB\Desktop\ir_new\ir-logmerge-starter\experiment-runs\6bbf617a87964f50acf64effb270bb1b
mode n flushes merges max_disk_indexes postings_written postings_written_with_finalize write_amplification seconds
---- - ------- ------ ---------------- --------------- ---------------------------- ------------------ -------
aux 50000 113 112 1 327308800 333021421 57.6432 27.6441
aux 100000 56 55 1 165003237 170762949 29.0592 16.1175
aux 200000 28 27 1 86578044 92335927 15.2475 10.1777
aux 400000 14 13 1 47192658 52949848 8.31122 6.5959
log 50000 113 109 6 36010697 41807682 6.34194 17.1865
log 100000 56 53 5 30381950 36141662 5.35064 12.2144
log 200000 28 25 4 24793384 30551267 4.36643 8.43963
log 400000 14 11 3 19197099 24954289 3.38085 5.89405
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter>
main* Bootstrap v5.3.8 RTL: N/A Kimi 0 0 Git Build Git Graph 127.0.0.1 world Connected to Discord
Ln 13, Col 46 Spaces: 4 UTF-8 CRLF PowerShell Go Live Prettier

```
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> Get-CimInstance Win32_Processor | Select-Object Name

Name
----
13th Gen Intel(R) Core(TM) i7-13620H

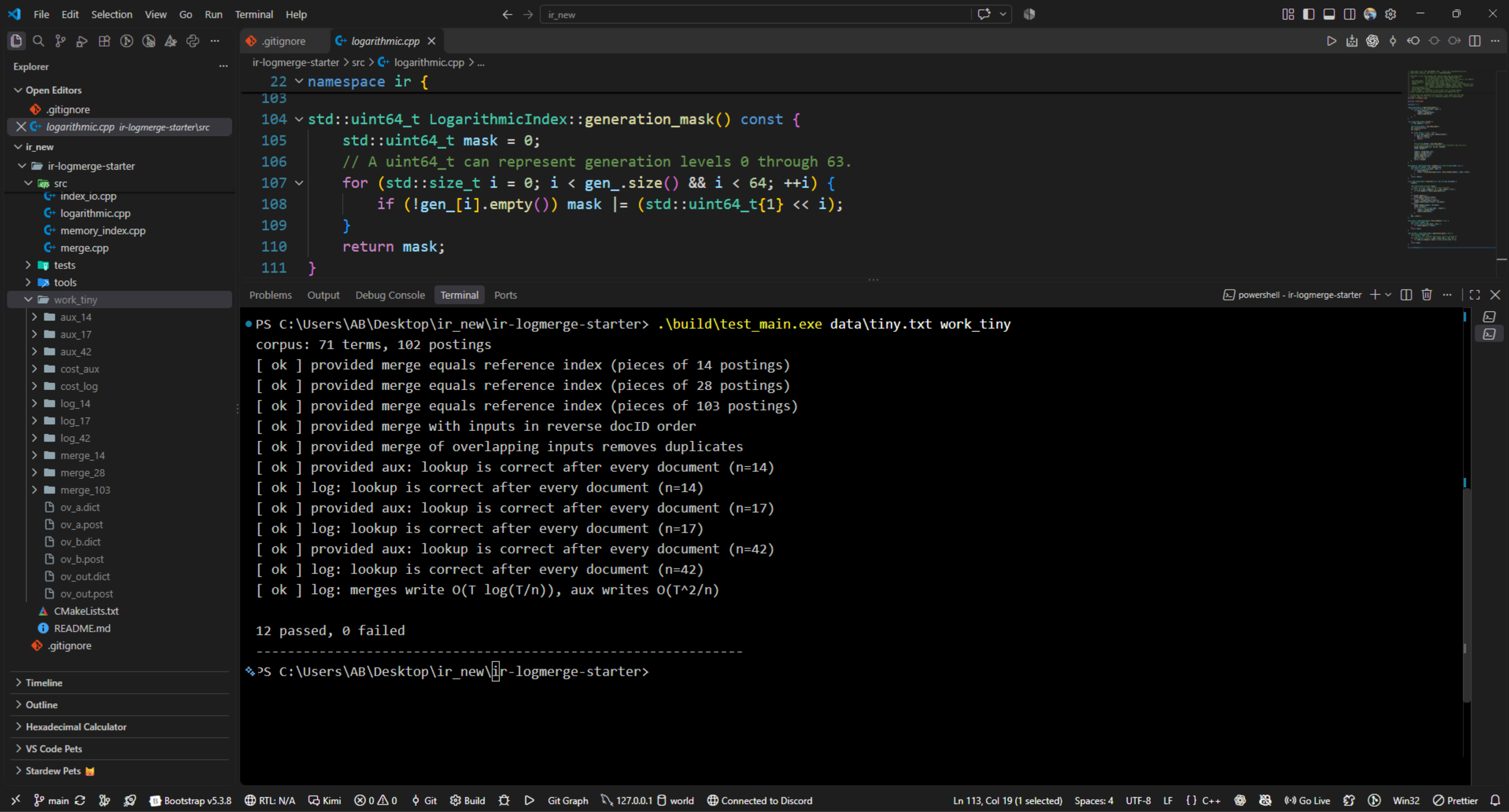


PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> Get-CimInstance Win32_ComputerSystem | Select-Object TotalPhysicalMemory

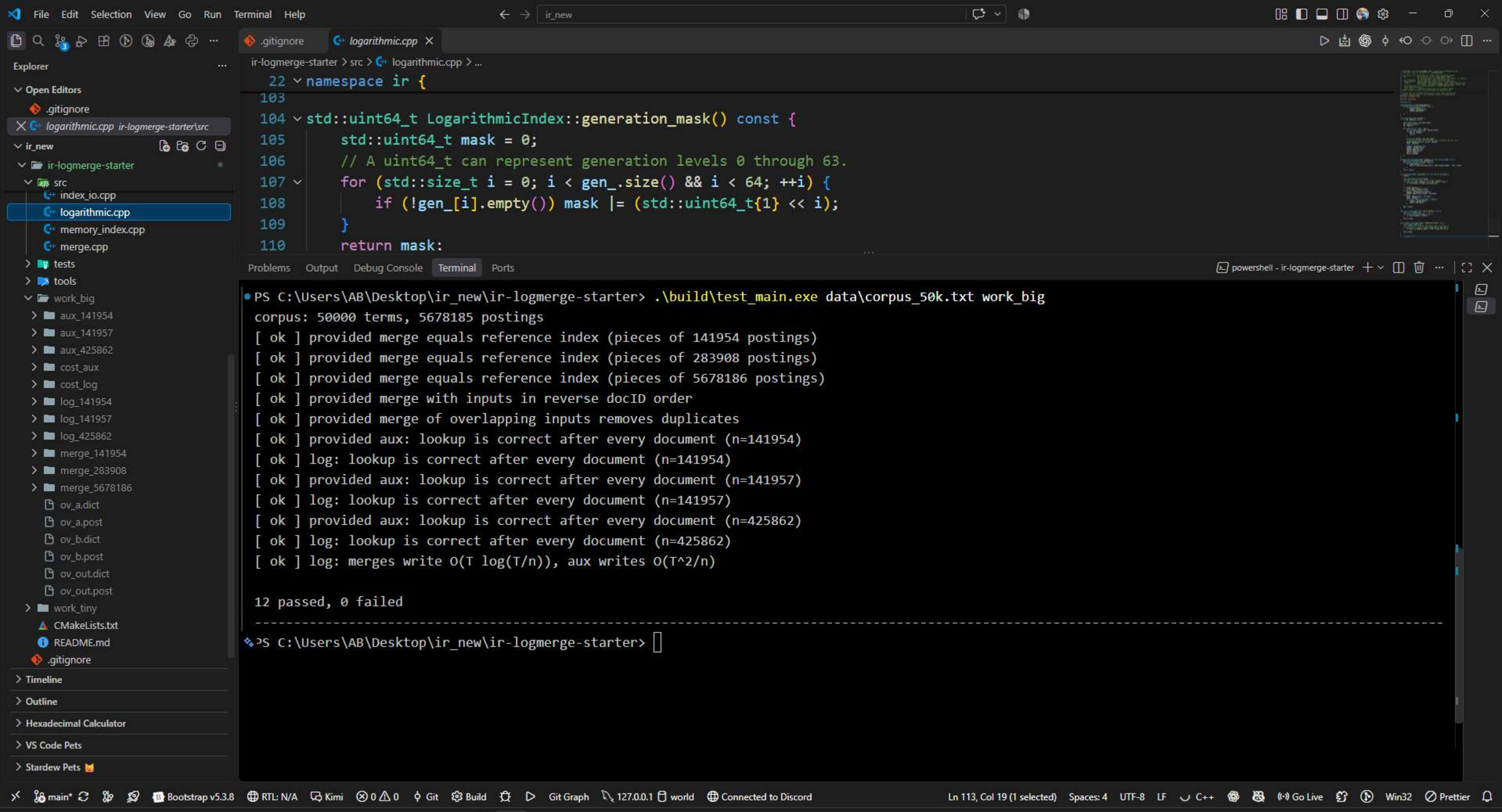
TotalPhysicalMemory
-------------------
        16788709376



PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> g++ --version
g++.exe (Rev2, Built by MSYS2 project) 14.2.0
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```


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ir_new
.gitignore
logarithmic.cpp
ir-logmerge-starter > src > logarithmic.cpp > ...
22 namespace ir {
103
104 std::uint64_t LogarithmicIndex::generation_mask() const {
105 std::uint64_t mask = 0;
106 // A uint64_t can represent generation levels 0 through 63.
107 for (std::size_t i = 0; i < gen_.size() && i < 64; ++i) {
108 if (!gen_[i].empty()) mask |= (std::uint64_t{1} << i);
109 }
110 return mask;
111 }
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logarithmic.cpp ir-logmerge-starter\src
ir_new
ir-logmerge-starter
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aux_17
aux_42
cost_aux
cost_log
log_14
log_17
log_42
merge_14
merge_28
merge_103
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ov_a.post
ov_b.dict
ov_b.post
ov_out.dict
ov_out.post
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Stardew Pets
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powershell - ir-logmerge-starter
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> .\build\test_main.exe data\tiny.txt work_tiny
corpus: 71 terms, 102 postings
[ok] provided merge equals reference index (pieces of 14 postings)
[ok] provided merge equals reference index (pieces of 28 postings)
[ok] provided merge equals reference index (pieces of 103 postings)
[ok] provided merge with inputs in reverse docID order
[ok] provided merge of overlapping inputs removes duplicates
[ok] provided aux: lookup is correct after every document (n=14)
[ok] log: lookup is correct after every document (n=14)
[ok] provided aux: lookup is correct after every document (n=17)
[ok] log: lookup is correct after every document (n=17)
[ok] provided aux: lookup is correct after every document (n=42)
[ok] log: lookup is correct after every document (n=42)
[ok] log: merges write O(T log(T/n)), aux writes O(T^2/n)
12 passed, 0 failed
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter>
main
Bootstrap v5.3.8
RTL: N/A
Kimi
0 0
Git
Build
Git Graph
127.0.0.1 world
Connected to Discord
Ln 113, Col 19 (1 selected)
Spaces: 4
UTF-8
LF
C++
Go Live
Win32
Prettier

```
------------------------------------------------------------
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> .\build\gen_corpus.exe data\corpus_50k.txt 50000
wrote 50000 documents to data\corpus_50k.txt
```


File Edit Selection View Go Run Terminal Help
ir_new
.gitignore
logarithmic.cpp
ir-logmerge-starter > src > logarithmic.cpp > ...
22 namespace ir {
103
104 std::uint64_t LogarithmicIndex::generation_mask() const {
105 std::uint64_t mask = 0;
106 // A uint64_t can represent generation levels 0 through 63.
107 for (std::size_t i = 0; i < gen_.size() && i < 64; ++i) {
108 if (!gen_[i].empty()) mask |= (std::uint64_t{1} << i);
109 }
110 return mask:
Explorer
Open Editors
.gitignore
logarithmic.cpp ir-logmerge-starter\src
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ir-logmerge-starter
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logarithmic.cpp
memory_index.cpp
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aux_141957
aux_425862
cost_aux
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merge_141954
merge_283908
merge_5678186
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ov_a.post
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powershell - ir-logmerge-starter
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter> .\build\test_main.exe data\corpus_50k.txt work_big
corpus: 50000 terms, 5678185 postings
[ok] provided merge equals reference index (pieces of 141954 postings)
[ok] provided merge equals reference index (pieces of 283908 postings)
[ok] provided merge equals reference index (pieces of 5678186 postings)
[ok] provided merge with inputs in reverse docID order
[ok] provided merge of overlapping inputs removes duplicates
[ok] provided aux: lookup is correct after every document (n=141954)
[ok] log: lookup is correct after every document (n=141954)
[ok] provided aux: lookup is correct after every document (n=141957)
[ok] log: lookup is correct after every document (n=141957)
[ok] provided aux: lookup is correct after every document (n=425862)
[ok] log: lookup is correct after every document (n=425862)
[ok] log: merges write O(T log(T/n)), aux writes O(T^2/n)
12 passed, 0 failed
PS C:\Users\AB\Desktop\ir_new\ir-logmerge-starter>
main* Bootstrap v5.3.8 RTL: N/A Kimi 0 0 Git Build Git Graph 127.0.0.1 world Connected to Discord
Ln 113, Col 19 (1 selected) Spaces: 4 UTF-8 LF C++ Go Live Win32 Prettier